

Cover Song Identification (CSI)

ORIGINAL TRACK

COVER TRACK



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Introduction to Cover Song Identification (CSI)

The Core Retrieval Challenge

Goal: Given a **cover** as query, retrieve the matching **original** from a gallery of tracks, based only on raw audio.



Timbral Variance

Instruments and vocals change completely between versions.



Tempo & Structure

Speed variations, added or removed sections, and structural shifts.



Key Transpositions

Uniform pitch shifting across the entire audio track.

Objective: Design a **style-invariant** representation capturing only **melody & harmony**.

Solution Overview

Bridging CSI to SSL & Metric Learning

Traditional Limitations:

Handcrafted features (Chroma) lack context, deep learning from scratch needs massive data (we have only 120 groups).



MERT Backbone

SSL backbone pretrained on 24,000 hours, separates timbre from melody across layers.



MLP Projector

Lightweight head trained via Metric Learning (Triplet/NT-Xent) → timbre-invariant embeddings.



DTW Matching

Temporal alignment handles tempo drifts & structural shifts.

Our Solution: MERT + Metric Learning Strategy

Dataset & Group-Aware Split

✓ Data Integrity & Split

Dataset:

240 tracks (120 original-cover pairs)

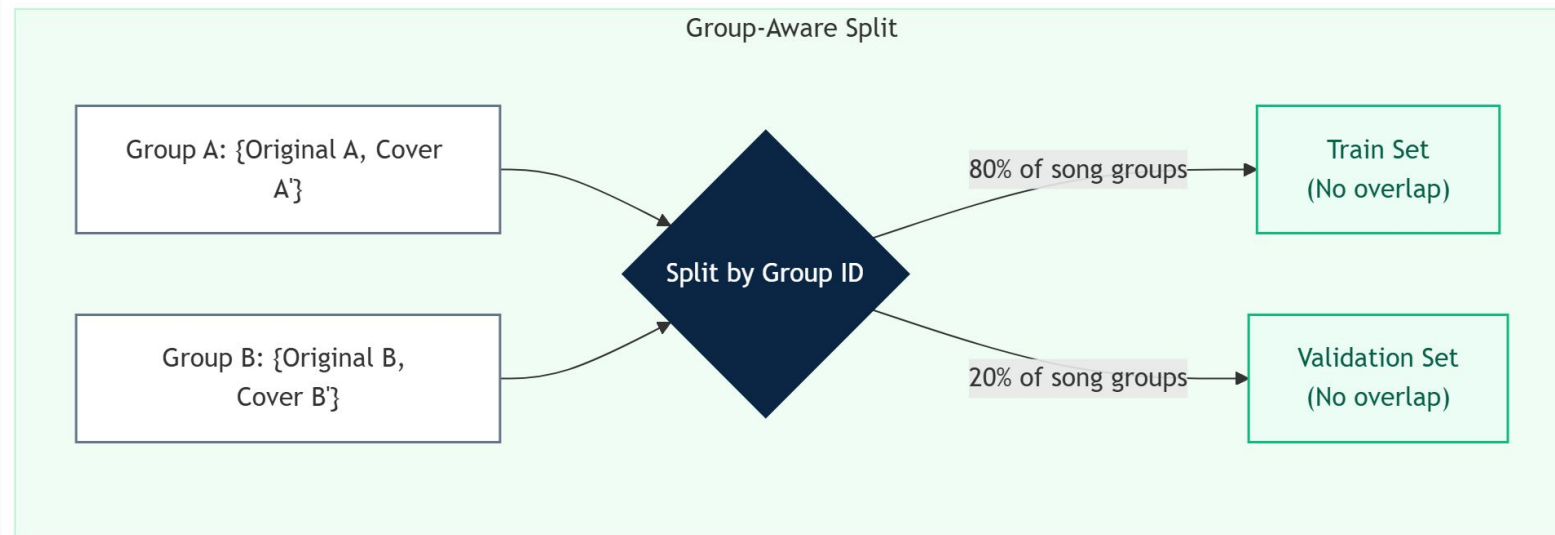
Retrieval Setup:

Query = cover Gallery = originals

Split Strategy:

By Group ID (Train: 96, Val: 24)

>> Zero Data Leakage Guaranteed



Modular System Architecture

High-Level Information Flow of the Proposed Solution



Training & Inference Strategy

- **Training:** 5-sec segments to maximize samples from 192 tracks.
- **Inference:** Processing whole tracks via DTW and pooling techniques.



Architecture Specifications

- **Backbone:** Frozen MERT-95M (95M parameters).
- **Projector:** Trainable MLP (~200K parameters).
- **Output:** 128-D embeddings normalized on a unit sphere.

System Core: Frozen SSL Backbone + Lightweight Trainable Head

Audio Segmentation & Preprocessing

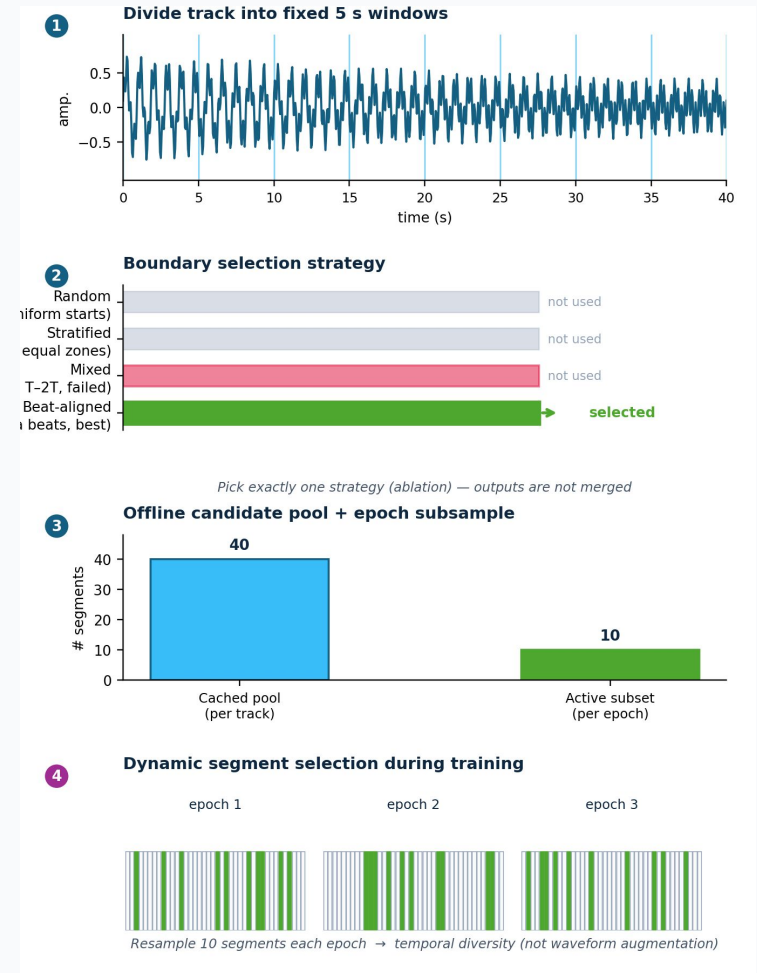
Dividing Tracks into 5-Second Chunks

Boundary Selection (4 strategies)

- **Random:** uniform random start positions
- **Stratified:** 10 equal zones, 1 segment each (uniform coverage)
- **Mixed:** variable length segments (T to 2T seconds) to force duration invariance (**failed**)
- **Beat-Aligned (Best):** snap to rhythm beats via `librosa` (reduces jitter)

Dynamic Segment Caching

Pool of 40 segments/track → sample 10 each epoch = **online temporal augmentation**



Trainable Projection Head Architecture

MLP Architecture (~200K parameters)

Purpose

Map high-dimensional MERT features ($x \in \mathbb{R}^{768}$) to compact embeddings ($z \in \mathbb{R}^{128}$) on the unit sphere.

Formula

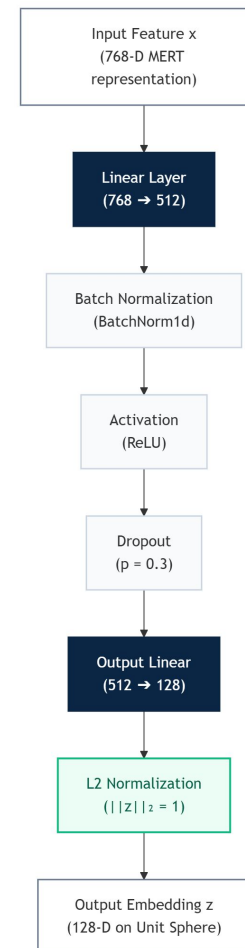
$$z = \text{L2-Norm}(W2 \cdot \text{Dropout}(\text{ReLU}(\text{BatchNorm}(W1 \cdot x))))$$

where $W1 \in \mathbb{R}^{768 \times 512}$ and $W2 \in \mathbb{R}^{512 \times 128}$

Distance

On unit sphere, cosine distance is equivalent to Euclidean distance:

$$d_{\cos}(z1, z2) = 1 - z1 \cdot z2 = \|z1 - z2\|_2^2 / 2$$



Deep Metric Learning & Loss Functions

Pull Covers Close, Push Unrelated Apart

Triplet Loss

Optimizes triplets:

- Anchor a (original track)
- Positive p (its cover)
- Negative n (unrelated song)

Hard Negative Mining

Selects a random positive, mines the **hardest negative** in batch.

NT-Xent Loss

Multi-positive InfoNCE loss. Positive cover as target, others as distractors.

4. Proxy-Anchor

(Best Silhouette)

Uses learnable proxy vectors for each song group instead of individual samples as anchors.



Evaluation Levels

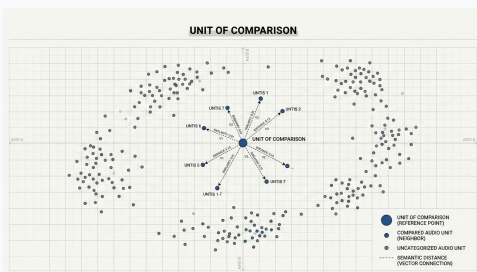
From Segment Embeddings to Track-Level Fusion

1. Segment-level

Unit: Individual Segment Embeddings

Each cover segment is compared against all original segments in the gallery.

- Baseline — no temporal or track-level fusion
- Risk: Many gallery segments per song → noisy ranking

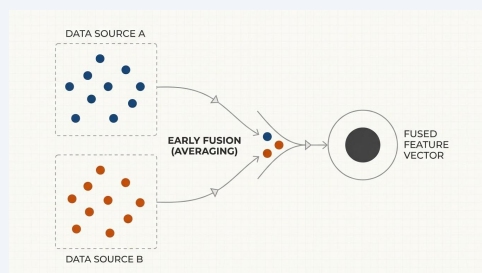


2. Track Pooling

Unit: Whole Tracks (Early Fusion)

All segment embeddings of a track are averaged into one track-level embedding.

- **Pros:** Simple, fast, robust aggregation
- **Cons:** Loses temporal order of segments



3. Majority Voting

Unit: Late Fusion

Match every cover segment to its best original segment (max cosine similarity).

- **Pros:** No temporal alignment, segments can match freely.
- **Metric:** Mean of best matches $\rightarrow$ MRR / Top-1 / Top-5.



3. Track DTW

Unit: Sequences (Late Fusion)

Aligned via Dynamic Time Warping (DTW) using cosine distance.

- **Pros:** Respects segment order; non-linear alignment
- **Cons:** Expensive; sensitive to sequence quality



Retrieval Metrics & Evaluation

Standardizing Performance Benchmarks

Retrieval Setup

Query: Cover segments/tracks

Gallery: Original segments/tracks ranked by similarity or DTW distance

Correct Hit: Original segment with the same **group_id**

MRR

Mean Reciprocal Rank

Calculates average $1/\text{rank}$ of the first correct hit.

Pros: Rewards early matches; smoother than Top-1.

Range: $0 \rightarrow 1$

TOP-1

Strict Accuracy

Fraction of queries where correct original is #1.

Cons: Ignores near-misses (rank 2 = 0).

Range: $0 \rightarrow 1$

TOP-5

Top-K Recall

Fraction where correct result appears in top 5.

Pros: More forgiving for noisy ranking environments.

Range: $0 \rightarrow 1$

SILHOUETTE

Cluster Quality

Measures embedding distance to same song vs. others.

Role: Auxiliary signal for training stability.

Range: $-1 \rightarrow 1$

Summary: Multi-metric evaluation ensures high-precision retrieval (Top-1) and robust embedding clustering (Silhouette).

Segmentation Results

Beat-Aligned + Dynamic Selection Wins

Variant	MRR (↑)	Δ MRR	Top-1	Top-5	Silhouette
Random (Base)	0.383	-	0.292	0.417	-0.080
Stratified	0.475	+0.092	0.375	0.583	-0.105
Mixed	0.323	-0.060	0.208	0.417	-0.163
Beat-Synchronized	0.478	+0.095	0.333	0.625	-0.011
Max Pool	0.320	-0.158	0.208	0.417	-0.055
Dynamic Pool (10/40)	0.475	+0.067	0.375	0.583	-0.105

Top Performer: Beat-Synchronized variant achieves the highest MRR (0.478) and Top-5 (0.625).

Loss Results

Variant	MRR (↑)	Δ MRR	Top-1	Top-5	Silhouette
Triplet Loss (Base)	0.422	-	0.292	0.500	0.019
NT-Xent (InfoNCE)	0.491	+0.069	0.375	0.625	0.021
Triplet Hard	0.519	+0.097	0.417	0.625	0.029
Proxy-Anchor	0.460	+0.038	0.333	0.625	0.048

Retrieval Matching Results

Track-level fusion strategies · Triplet Hard + GPS + BN · same trained embeddings

Variant	MRR (↑)	Δ MRR	Top-1	Top-5	Silhouette
Segment-level (baseline)	0.265	-	0.163	0.367	0.029
Track pooling (early fusion)	0.362	+0.097	0.208	0.583	0.029
Majority voting (late fusion)	0.470	+0.205	0.333	0.583	0.029
Track DTW (sequence alignment)	0.519	+0.254	0.417	0.625	0.029

Top Performer: Track DTW (Sequence Alignment) achieves the highest MRR (0.519) and Top-5 (0.625).

Embedding Collapse & Regularization

Preventing Trivial Solutions

THE CHALLENGE

Dimensional Collapse

All embeddings collapse to a single point.
Results in $\text{loss}=0$ but zero discrimination.

Early Stopping

Patience-based (12 epochs). Metrics correspond to the **best checkpoint**.

PROPOSED SOLUTIONS

GroupPairBatchSampler

Guarantees positive pairs per batch to ensure valid training signal.

BatchNorm

Zero mean, unit variance before L2 normalization. Effectively disperses points on the unit sphere.

Feature Augmentation

Injects Gaussian noise ($\sigma=0.02$) and applies dropout ($p=0.3$) on cached features.

Standard Random Sampler (No matching pairs)

Batch: [S1_Orig, S5_Cover,
S8_Orig, S12_Cover, ...]

GroupPairBatchSampler (Guaranteed Positive Pairs)

Pair 1: Original + Cover

Pair 2: Original + Cover

Batch: [S1_Orig, S1_Cover,
S5_Orig, S5_Cover, ...]

Objective Alignment

By combining strategic sampling, batch normalization, and feature loading regularizations, the model avoids trivial convergence and maintains discriminative power across the embedding space.

Sampling & Batch Normalization

Optimizing Training Signals and Embedding Stability

GROUPPAIRBATCHSAMPLER

Efficient Positive-Pair Retrieval

Guarantees every song group has both **Original** and **Cover** segments together in each training batch.

Batch Composition

G song groups $\times$ S segments per role

$2 \times G \times S = \text{Batch Size (e.g., 32)}$

Why It Matters

Random sampling often misses cover-original pairs, yielding weak training signals. This sampler enables effective hard-negative mining.

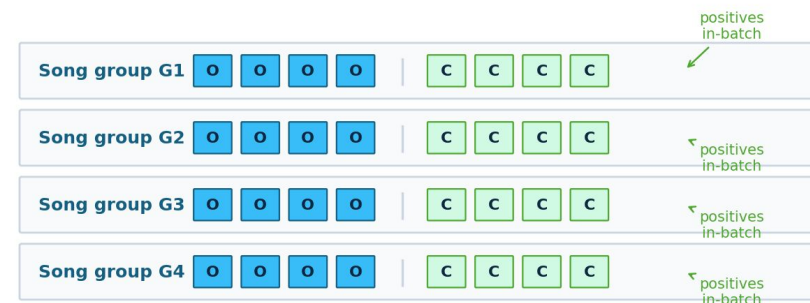
PROJECTION HEAD

BatchNorm & Normalization

Linear $\rightarrow$ BatchNorm1d $\rightarrow$ ReLU $\rightarrow$ Dropout $\rightarrow$ Linear $\rightarrow$ L2

GroupPairBatchSampler — one training batch

$2 \times G \times S = \text{batch size}$ (e.g. $G=4$ groups, $S=4$ segments $\rightarrow$ 32)



O = original segments C = cover segments (same song group)

vs. shuffle: cover-original pairs often land in different batches

Ablation results show consistent performance gains when combining group-based batching with BatchNorm layers.

Data Augmentation — Types and Where They Apply

Pipeline: Audio →  **Segments** → Feature Extraction (MERT) → Cache →  **Training (Projection Head)** → Evaluation
Aug 1 (Offline) Aug 2 (Online)

1. Waveform Augmentation (Offline)

Time / Waveform

What: Random transforms on raw waveform before MERT

Transforms: Pitch shift (± 2 semitones), time stretch (0.92–1.08 $\times$), gain, Gaussian noise, low/high shelf EQ

Stage: Feature extraction only (extract_features.py)

When: Once, before training — not re-applied each epoch

Cached? Yes — augmented embeddings stored in features.pt

Coupling: Same seed → identical transform on cover and original

2. Feature-level Augmentation (Online)

Training-Only Regularization

What: Perturbations on cached MERT embeddings inside head

Transforms: Feature dropout (10% on input) + Gaussian feature noise ($\sigma = 0.02$)

Stage: Training only (ProjectionHead.forward)

When: Every training batch — new noise/dropout each forward

Cached? No — applied on the fly

Eval: Disabled — clean embeddings at val/test

Summary: Dual-layer strategy ensuring data diversity (Offline) and model robustness (Online).

Sequential Ablation Results

Triplet Hard · beat segments · shuffle-batch baseline · track-DTW evaluation

Variant	MRR (↑)	Δ MRR	Top-1	Top-5	Silhouette
Baseline	0.457	-	0.333	0.542	0.004
GroupPairBatchSampler	0.538	+0.081	0.458	0.583	-0.014
BatchNorm	0.519	+0.062	0.417	0.625	0.029
Feature-level aug	0.495	+0.038	0.417	0.542	0.031

Top Performer: Group Pair Batch Sampler achieves the highest MRR (0.538, Δ +0.081 vs baseline).

Augmentation Comparison Results

Time-domain (waveform) vs feature-level (cached embeddings) augmentation

Variant	MRR (↑)	Δ MRR	Top-1	Top-5	Silhouette
Baseline (shuffle)	0.457	-	0.333	0.542	0.004
Time-domain aug	0.361	-0.096	0.208	0.542	-0.145
Baseline (GPS + BN)	0.519	-	0.417	0.625	0.029
Feature-level aug	0.495	-0.024	0.417	0.542	0.031

Time-domain aug: Δ MRR -0.096 vs shuffle baseline. Feature-level aug: best silhouette (+0.031) but Δ MRR -0.024 vs GPS+BN.

MERT Layer Hierarchy

How Different Layers Capture Musical Information

Low Layers (0-3)

Timbre & Acoustics

Considered **Noise** for CSI as instruments/vocals change completely in covers.

Mid Layers (4-8)

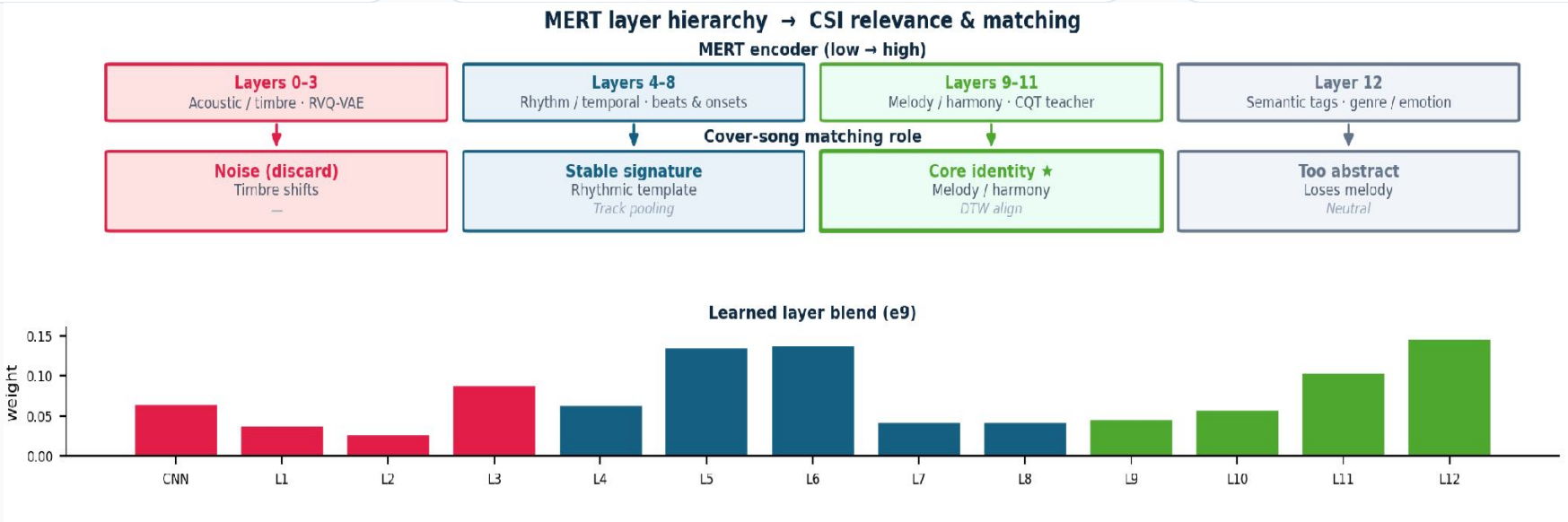
Rhythm & Beats

Represents the **Stable Signature** of a track, utilized for Track Pooling.

High Layers (9-11)

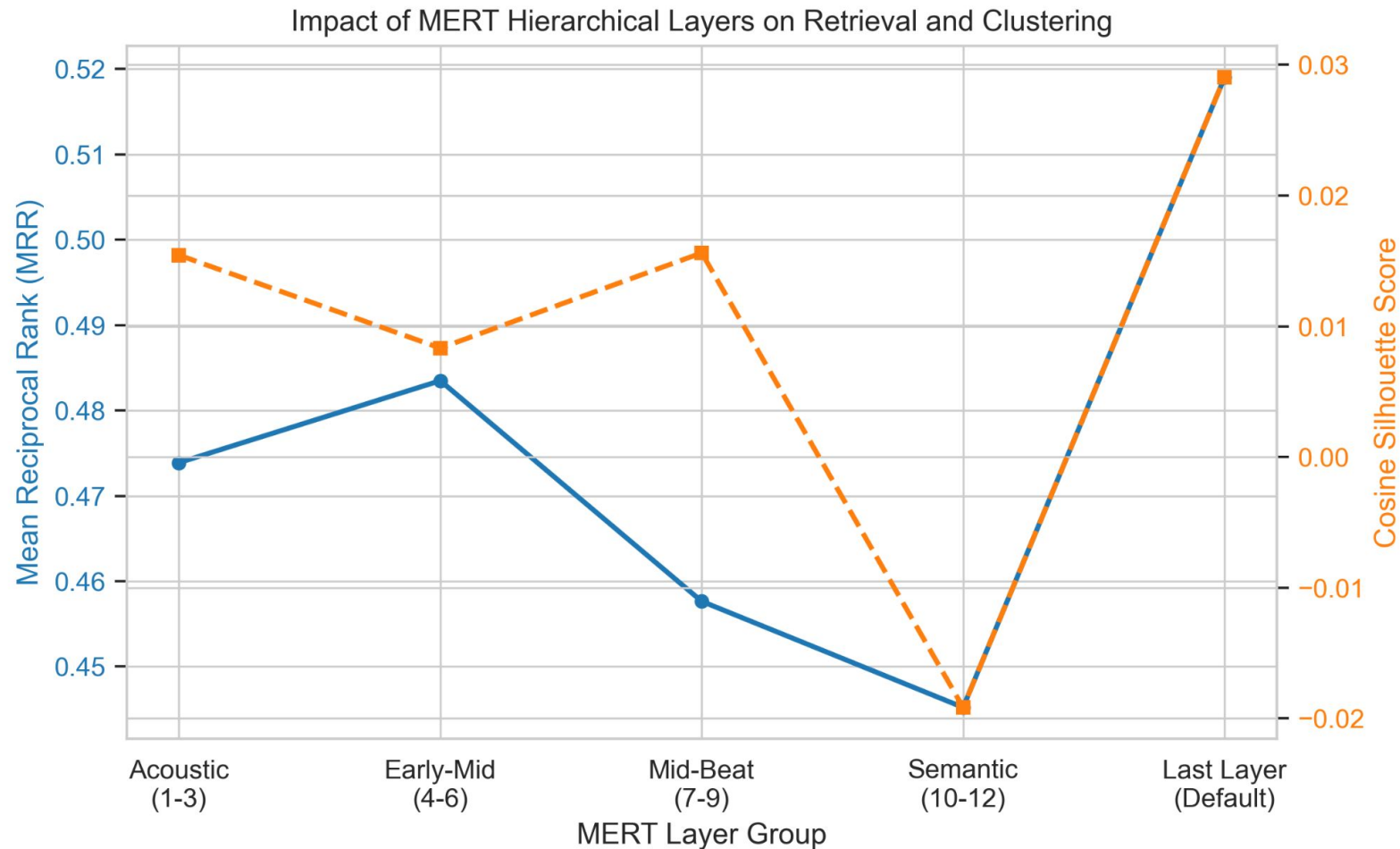
Melody & Harmony

Defines **Core Song Identity**, enabling effective DTW alignment.



MERT Layer Pooling Results

Which Layers Capture Music Semantics?



Conclusion & Optimal Configurations

Optimization Goal	Winning Configuration	Metric Score
Best Absolute Ranking	Triplet Hard (Without BN)	MRR: 0.538
Best Top-5 Match	Triplet Hard (With BN)	Top-5: 0.625
Best Cluster Stability	Proxy-Anchor	Silhouette: +0.048
Best Efficiency vs DTW	Early-Mid Layers (Mean Pool)	Pool MRR: 0.510

The Efficiency Anomaly: Early-Mid Layers + Mean Pool ($O(1)$) matches rigorous DTW alignment ($O(N^2)$) by naturally isolating melodic semantics.

Key Takeaway: Deep Metric Learning successfully embeds temporal melody into scalable, static 128-D vectors.

Thank you

Questions & Discussion